

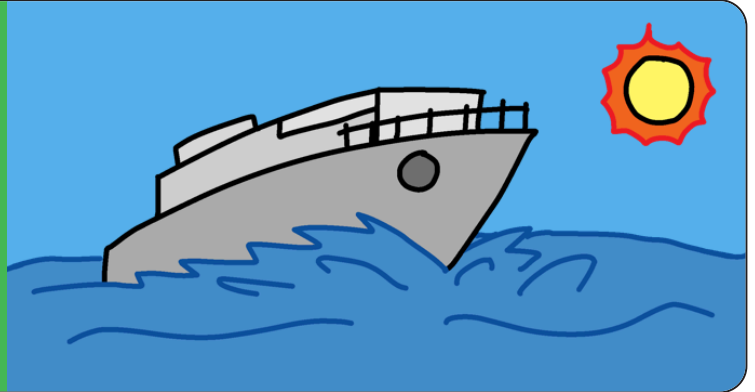
Projects

Boat race

the-cc.io/boat-race-editor

Make a boat racing game!

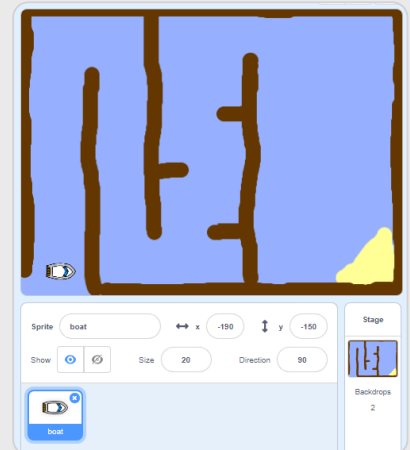
Use the mouse to steer a boat to an island without bumping into obstacles.



Starter project

This project starts from a starter project with the boat sprite and a race course to steer around.

- **In the Code Editor:** go to the-cc.io/boat-race-editor. The starter project opens with the instructions.
- **In Scratch:** open rpf.io/boat-race-starter-on. If you have a Scratch account, click **Remix** to make your own copy.

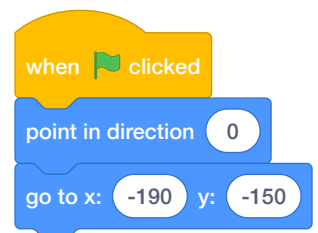


1 Starting the boat

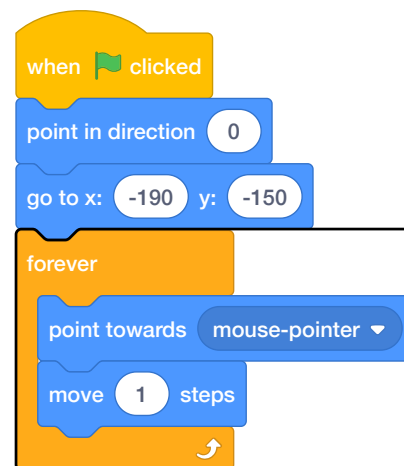


Make the boat start in the corner and follow the mouse pointer.

- ☐ Click on the **Boat sprite**, then add these blocks to make a starting position.



- ☐ Add **forever** and **point towards** blocks to **move** while following the cursor.



- ☐ **Now run your code** Click the green flag and move your mouse.

Check that the boat follows the pointer around the screen.

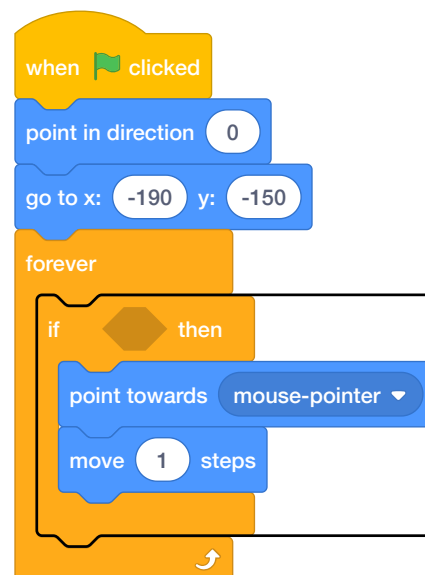


2 Smooth out the movement

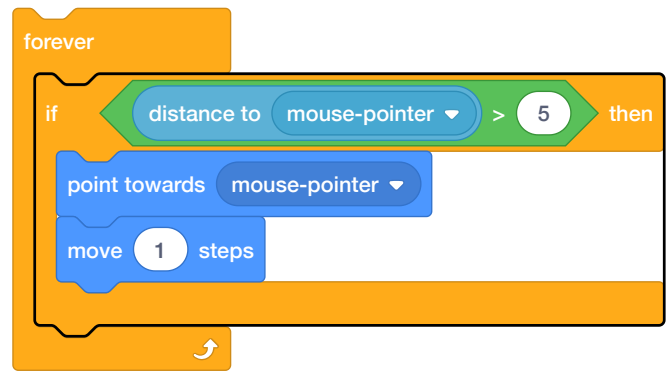


Make the boat stop moving when it touches the pointer.

- ☐ Wrap the movement in an **if** block.



- ☐ Add a **more than** block and change the number to **5**. Drag a **distance to mouse-pointer** block into the other slot.



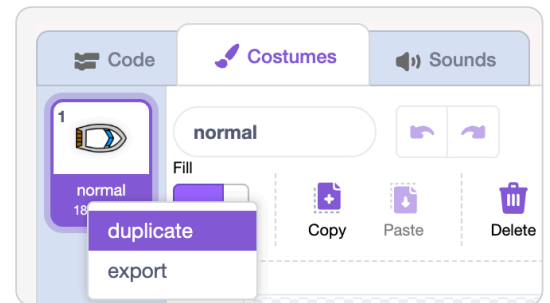
- ☐ **Now run your code** Click the green flag and move your mouse near the boat. Check that the boat stops jittering.

3 Crash costumes

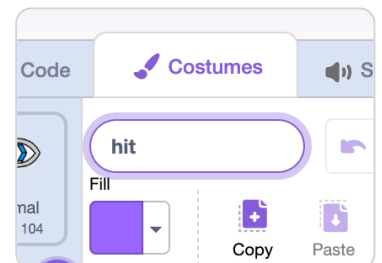


Give the boat a second costume that looks like it is wrecked.

- ☐ In the **Costumes** tab, duplicate the costume.

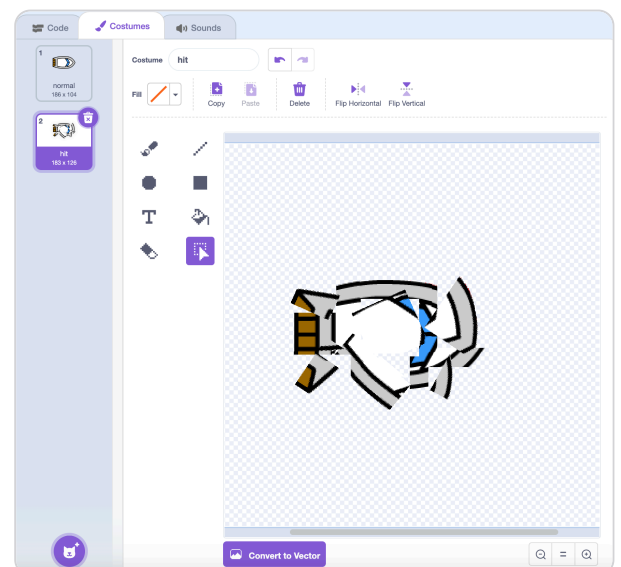


- ☐ Name the new costume **hit**.



- ☐ In your **hit** costume use the paint editor tools to move and rotate parts of the boat to make it look smashed up.

Check that you now have two costumes, **normal** and **hit**, and that they look different.

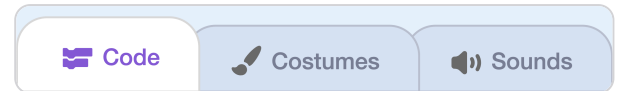


4 Detect a crash

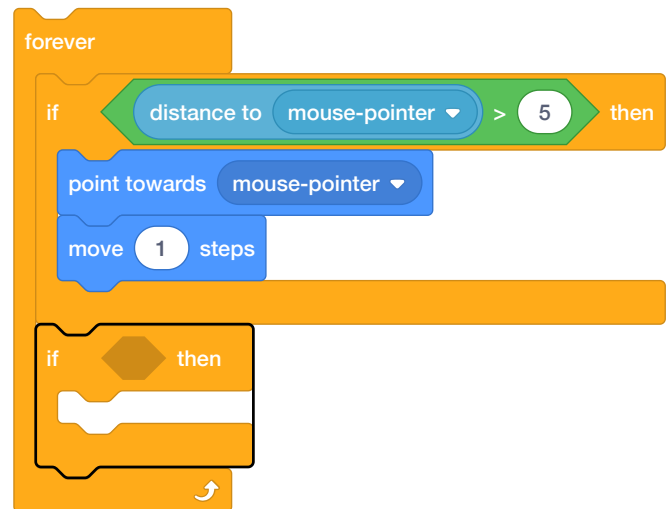


Make the boat crash when it touches the wooden barrier.

☐ Click on the **Code** tab.



☐ Inside the **forever** loop, add an **if** block.



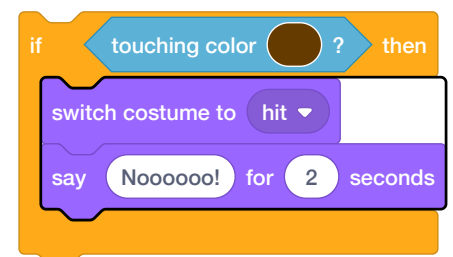
☐ Add a **touching color** block to detect if the boat is touching brown wood colour.



Use the eyedropper tool to set the same brown as the wood.



☐ If touching the wood, **switch costume** to hit and **say** "Noooooo!".



☐ **Now run your code** Click the green flag and crash into a barrier.

Check that the boat shows its wrecked costume and says Noooooo!



5 Reset after a crash



- ☐ Send the boat back to the start when it crashes.

Inside the **if** block, add **movement** and **costume** blocks to reset the boat.

- ☐ **Now run your code** Click the green flag.

Crash into a barrier and check that the boat goes back to the start.



6 Start looking normal

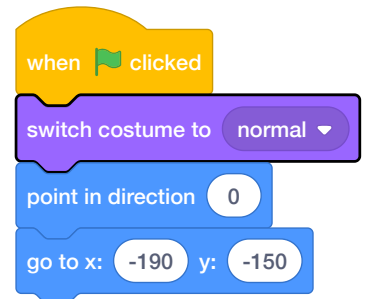


- ☐ The boat needs to start each game on its **normal** costume.

Add a **switch costume to** block right after **green flag**.

- ☐ **Now run your code** Crash the boat, then click the green flag.

Check that the boat starts on its **normal** costume.

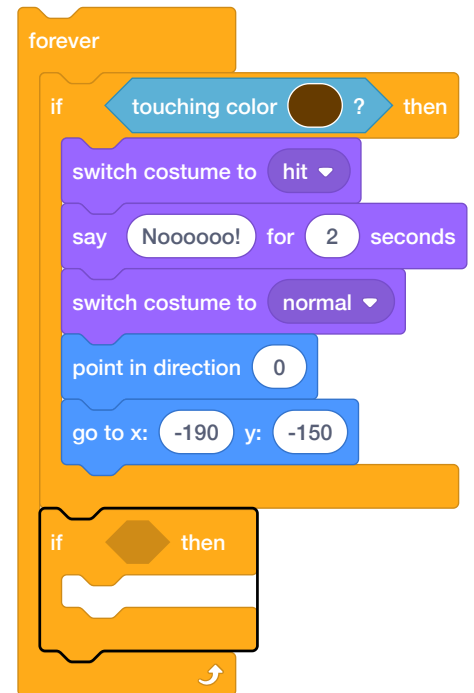


7 Winning

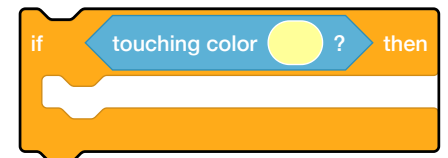


The player wins when the boat reaches the island.

- ☐ At the bottom of the **forever** loop, add an empty **if** block.



- ☐ Add a **touching color** block and use the eyedropper tool to select the yellow colour of the island.



- ☐ Add a **say** YEAH! for when you win and **stop all**.

Tip

To test winning quickly, change the first **go to** block to **go to x: (150) y: (-90)**.

Change it back to **go to x: (-190) y: (-150)** afterwards.



- ☐ **Now run your code** Click the green flag and steer to the island.

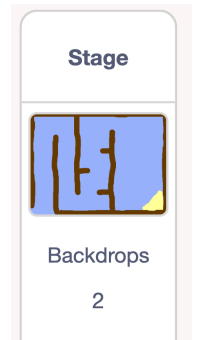
Check that the boat says YEAH! and the game stops.

8 Add a timer



Add a timer so the player has to reach the island quickly.

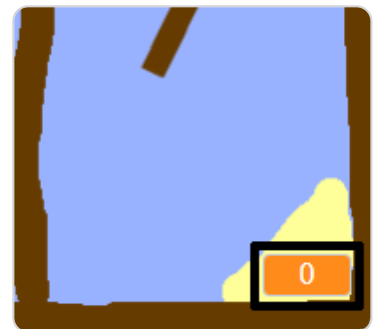
- ☐ Click on the **Stage**.



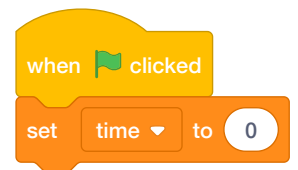
- ☐ In the **variables** menu, make a new variable and name it **time**.

Make a Variable

An orange box will appear on the stage.

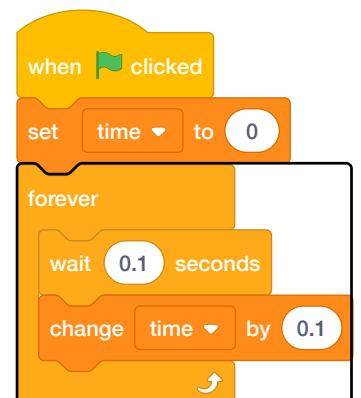


- ☐ Add a **green flag** and **set** the time to **0**.



- ☐ Add a loop that changes the time every **0.1** seconds.

- ☐ **Now run your code** Click the green flag and check that the timer counts up as you play.



9 Booster arrows

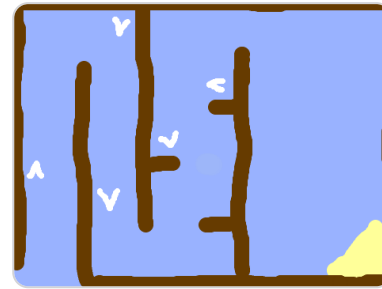
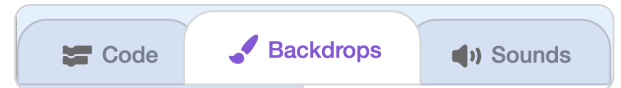


Add boosters to speed the boat up.

- ☐ On the **Stage**, click on the **Backdrops** tab and paint some white booster arrows onto the backdrop.

Tip

The project includes a ready-made backdrop with arrows, if you'd rather use that than paint your own.



The arrows do nothing until the next step.

Check that white arrows now appear on your backdrop.

10 Speed the boat up



Make the boat go faster over a white arrow.

- ☐ On the **Boat sprite**, inside the **forever** loop and after the winning check, add an **if** and **touching color** block. Use the eyedropper tool to set the colour to the white arrows.



- ☐ Move the boat faster by adding a **move** block and changing it to **3**.



- ☐ **Now run your code** Click the green flag and drive over an arrow.

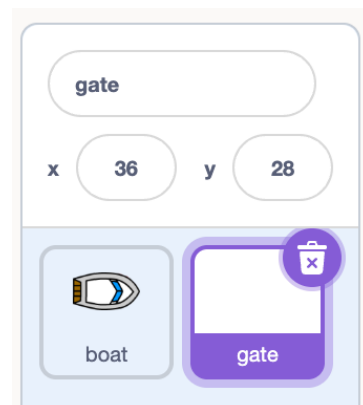
Check that the boat speeds up.

11 Make a spinning gate



Add an obstacle for the boat to avoid.

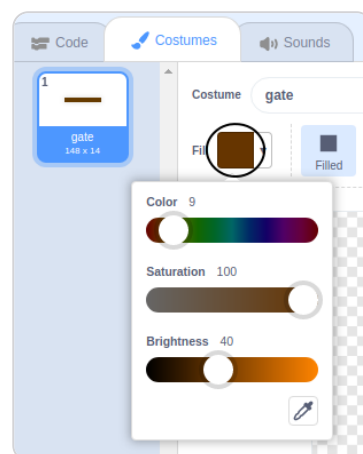
- ☐ Paint a new sprite and name it **gate**.



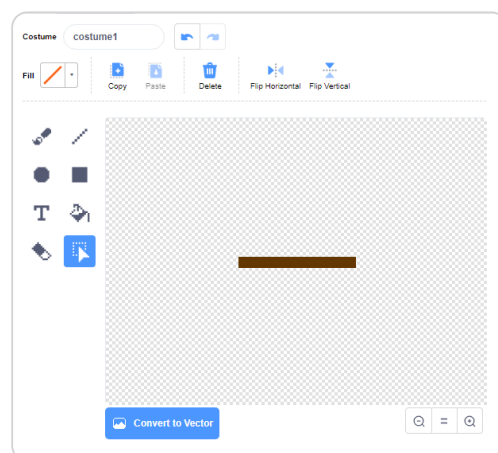
- ☐ Match the fill colour to the wooden barriers.

Tip

If the colour is hard to match, set it to colour 9, saturation 100, brightness 40.

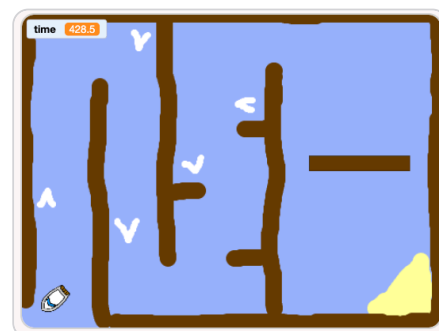


- ☐ Draw a gate with the Rectangle tool. Centre the gate in the middle of the costume canvas, so it spins on the spot.



- ☐ On the stage, drag the gate into the boat's path, between the start and the island.

Resize it if it is too big. Change the size using the size box under the stage.

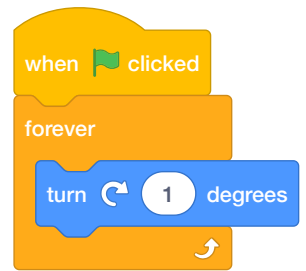


12 Spin the gate



- ☐ Make the gate spin round so it is harder to steer past.

In the **Code tab**, add a **forever** loop with a **turn 1 degree**.



- ☐ **Now run your code** Click the green flag.

Check that the gate spins and that hitting it makes the boat crash.



13 Challenge

Here are some ideas to make the game your own.

Pick one idea, make the change, then test it.

1 Add sound effects

You can add **sounds** that play when the boat crashes or reaches the island.

You could also add background music to the Stage.

2 Add more obstacles

For example, add green slime to your backdrop and change the code so touching it slows the boat down.

You could add a moving obstacle, like a log or a shark!



3 Turn your game into a race between two players

The second player could control their boat with the up arrow key to move forward and the left and right arrow keys to turn.

4 Create more levels by adding different backdrops

Can you then let the player choose between levels?